“Nonlinear 2-D Tank Filling Model using Finite Difference Method and Machine Learning”

Introduction

Tank filling is a common process in industries such as water storage, fuel tanks, and chemical plants. During filling, the water surface does not rise smoothly but shows oscillations due to inertia and gravity. For small motions, the behavior is linear, but for large motions, nonlinear effects become important. Therefore, a nonlinear model is required to study the system accurately.

Why it is required?

- It Simplifies Complex Physics
- Capture Important Behavior(water level rise, surface oscillation, over flow condition)
- Used in Engineering Design
  - Start with 2-D model
  - Validate concept
  - Then move to 3-D CFD simulation

➢ Real-Life Where 2-D Approximation Works Well
  - Long rectangular tanks
  - Channels and pipelines
  - Narrow tanks

Problem Description

Consider a rectangular 2-D tank being filled with water from an inlet. As water enters:

The water level increases

Surface oscillations are generated

Large filling rates produce nonlinear behavior

The objective is to predict the variation of water surface height with time and analyze the effect of nonlinearity.

Mathematical Model

The system is modeled using the Forced Damped Duffing Equation:

$$\eta'' + cn' + w^2\eta + \beta\eta^3 = Q(t)$$

Parameter	Description
n(t)	Water surface displacement
n'(t)	Velocity of surface
n''(t)	Acceleration
C	Damping coefficient
ω	Natural frequency
β	Nonlinear parameter
Q(t)	Inlet flow rate

Assumptions

1. Flow is two-dimensional
2. Fluid is incompressible
3. Tank is rectangular
4. Surface motion is represented by a single dominant mode

Finite Difference Method

$$\eta'' = \frac{\eta_{n+1} - 2\eta_n + \eta_{n-1}}{h^2}$$

$$\eta' = \frac{\eta_n - \eta_{n-1}}{h}$$

Final Discretized Equation

$$\eta_{n+1} = 2\eta_n - \eta_{n-1} - h^2\left[c\left(\frac{\eta_n - \eta_{n-1}}{h}\right) + \omega^2\eta_n + \beta\eta_n^3 - Q\right]$$

Numerical Problem :-

Given

-) $c = 0.5$

$\omega = 3$

$\beta = 60$

$Q = 0.7$

$h = 0.1$

initial Condition

$\eta(0) = 0.05$

$\eta'(0) = 0$

Matlab Code for Finite Difference Method Solution Variation of Water Level With Time

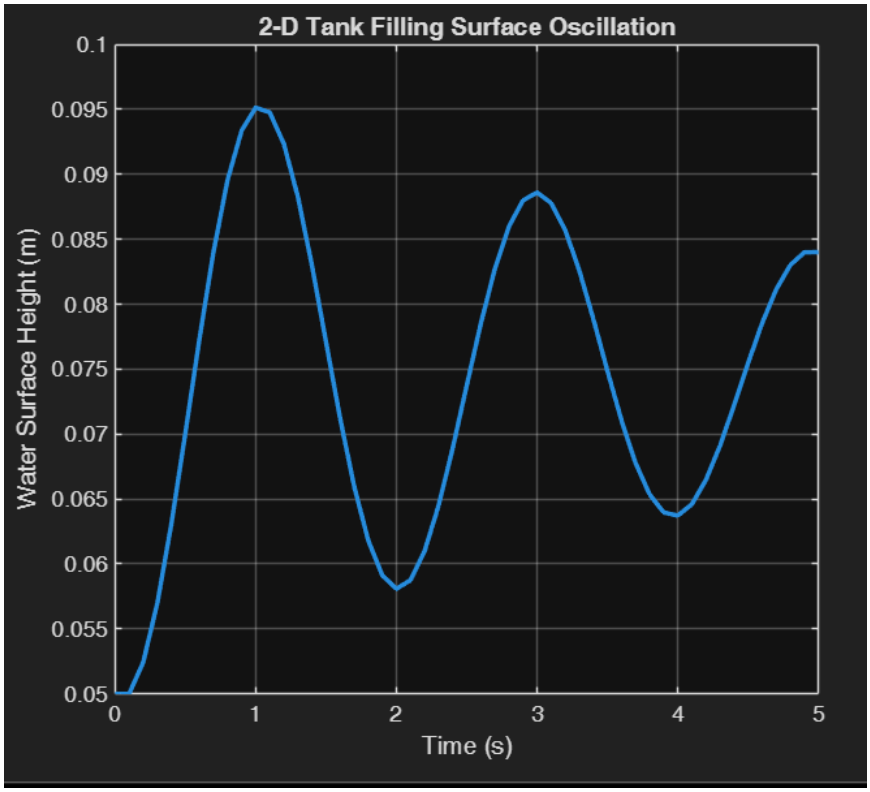
```
clc

clear

close all
```

```
% parameters
c = 0.4;
omega = 3;
beta = 60;
Q = 0.7;
h = 0.1;
T = 5;
t = 0:h:T;
N = length(t);
eta = zeros(1,N);
% initial conditions
eta(1) = 0.05;
eta(2) = 0.05;
for n = 2:N-1
eta(n+1) = 2*eta(n) - eta(n-1) ...
- h^2*( c*(eta(n)-eta(n-1))/h ...
+ omega^2*eta(n) ...
+ beta*eta(n)^3 ...
- Q );
End
plot(t,eta,'LineWidth',2)
xlabel('Time (s)')
ylabel('Water Surface Height (m)')
title('2-D Tank Filling Surface Oscillation')
grid on
```

Output



Machine Learning Code Using Python For Random Time Prediction

```
import numpy as np

# generate data

np.random.seed(42)

samples = 200

c = np.random.uniform(0.2,1.0,samples)

omega = np.random.uniform(2,6,samples)

beta = np.random.uniform(20,150,samples)

Q = np.random.uniform(0.5,1.5,samples)

# target (approx relation)

y = 0.05 + 0.08*Q - 0.02*c + 0.0008*beta + 0.01*omega

# input matrix

X = np.column_stack((c, omega, beta, Q))
```

```
# add bias term
X = np.hstack((np.ones((samples,1)), X))
# normal equation (linear regression)
theta = np.linalg.inv(X.T @ X) @ X.T @ y
# prediction function
def predict(c, omega, beta, Q):
    x = np.array([1, c, omega, beta, Q])
    return x @ theta
# test
print("Predicted height:", predict(0.4,3,60,0.7))
```

Output

Predicted height: 0.1759999999999989

Conclusion

The nonlinear 2-D tank filling model was successfully analyzed using the finite difference method. The results show that nonlinear effects significantly influence surface motion. Machine learning helps in fast prediction and improves real-time system performance.